Atlas of Sustainability

2025 Design Challenge Organizers

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Introduction:
Atlas of Sustainability...in a Week?

Robert E. Roth, the University of Wisconsin Cartography Lab

The ***University of Wisconsin Cartography Lab (Cart Lab) Design Challenge*** is a day-long mapping workshop that brings together Cartography and Geography students, often in collaboration with campus or community partners, around a curated mapping theme. The Design Challenge is now an institution in the University of Wisconsin–Madison Geography Department, first running in 2015 and wrapping its 10th annual event this past February 2024. This year (officially #DC11) we brought the Design Challenge to an international stage, combining efforts with the ***Technical University of Wien Cartography Research Unit***, the ***Erasmus Mundus MSc Cartography Programme***, and the ***International Cartographic Association Commission on the User Experience (UX)***.

Specifically, our respective groups came together to make a collective atlas that tells regional and local stories about the United Nations Sustainable Development Goals (SDGs) ***... in just one week!*** For background, the United Nations adopted the 2030 Agenda for Sustainable Development in 2015, an expansion from the prior 2000–2015 Millennium Development Goals, and defined a global indicator framework for collective transformation and action toward a more equitable and just world. At the time of this mapping workshop (May 2025), the SDG global indicator framework comprised 17 goals to address the most pressing problems facing our worlds, 169 targets that prescribe real and actionable outcomes for sustainability, and 231 unique indicators used to measure and monitor progress towards these targets, inform policy at global and local levels, and promote accountability across all stakeholders (https://unstats.un.org/sdgs/).

The 2025 event, held in Vienna May 12-16, built on prior collaborations among the International Cartographic Association, the United Nations, and the UW Cart Lab to develop the ***Mapping for a Sustainable World*** (Kraak et al. 2020) open source Cartography textbook and subsequent spin-off activities to create the ***QGIS Technical Supplement*** (Houtman et al. 2021) supporting mapping of the SDG global indicator framework. The 2025 event was an overwhelming success, demonstrating the promise and possibilities of international collaboration. The resulting ***Atlas of Sustainability*** follows the ***Guerilla Cartography*** "Atlas" model (Cowart et al. 2025) and includes 17 two-page spreads, each covering an SDG, plus some front and back matter. Each spread includes one page for maps, charts, and clarifying annotation about the selected SDG and a second page providing a title and statement about what the maps are saying and what this means for our planet's future and our individual actions.

MSc Cartography students along with visitors from Masaryk University (Brno) worked in small teams to develop one of the atlas spreads. As part of mapping SDG indicator data, we also asked the international students to bring their own personal experiences and home locations into the map, resulting in 17 compelling stories grounded from situated perspectives to explore the alternatives that are possible from their different, worldwide perspectives. The #DC11 schedule was broken into six 90-120 minute units spanning three days, with most scheduled activities on Monday, May 12th, and drafts submitted 24 hours after the final unit on Thursday, May 15th. We aseembled in its presented form on Friday, May 16th. As an ongoing project, we invite you all to join us in mapping for a sustainable world...the atlas.

What's Beneath the Poverty Line?

Federica Ballardini & Thijs van Laar

We approached ***Sustainable Development Goal 1 on No Poverty*** with the awareness that poverty is a topic very much at the forefront of discussions on global development. It is maybe one of the most important questions when we talk about global development, and we really wanted to address it from a standpoint of trying not to fall into the typical narratives. We tried to not just thinking "inside the box" and work within the existing global economic system, but instead tackle the system itself. A lot of the time, the goals that we have as communities, as countries, and—especially—as international organizations fail to really address the root of the problem: we live in a system that is still embedded in colonial, extractivist, and hierarchical practices of production and consumption. If we do not address these issues first, every type of solution will be a plaster, a very superficial fix. Our approach was to stay critical, remaining aware of the structural dimensions of poverty and understanding that development cannot just be about alleviating symptoms—it has to be about confronting power imbalances and systemic injustice. Otherwise, we risk reinforcing the very inequalities we claim to be fighting against.

The design process was very much limited by the availability of data, which in many cases—especially for UN SDG-specific data—was quite incomplete. Globally and regionally, we struggled to find a location with data that was both comprehensive and relevant, which meant we had to work around this issue of data incompleteness for quite some time. Eventually, we found more complete data sources, particularly poverty data from the World Bank and additional academic research data, which then allowed us to define a clearer direction for our project. Our design process was therefore initially guided by data availability rather than starting with a fixed narrative or region. Once we had the data, we selected a region that not only had better data coverage but also aligned with our broader message: to challenge the dominant narrative that equates poverty only with the Global South or Africa. We wanted to dig deeper, going beyond the surface-level idea of the poverty line to explore structural causes. After this conceptual work, we focused on design elements—selecting an effective color palette and working to refine the visual clarity of our multivariate representation using a choropleth map beneath a qualitative point symbol map to compare poverty to the extraction of natural resources.

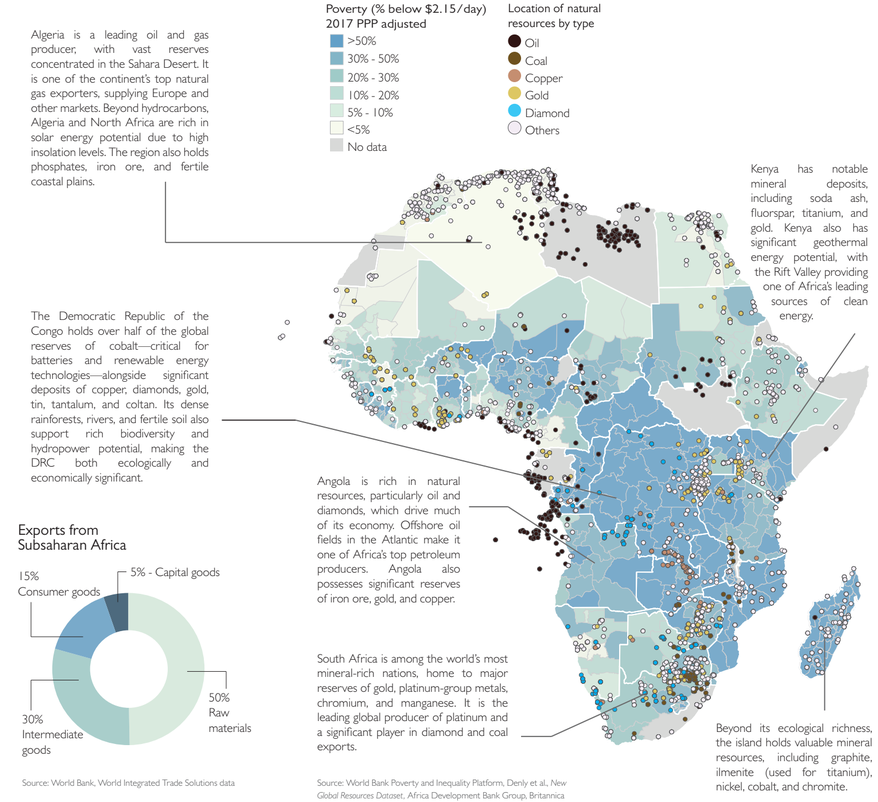
In our motivational approach, we wanted to focus on the idea of poverty from both a traditional and innovative standpoint. We chose to include the most conventional indicator—extreme poverty as defined by the international $2.50 a day threshold. This allowed us to visually present poverty in a familiar, recognizable way. However, we also aimed to go beneath and beyond this poverty line, questioning what poverty really means and how it is defined. Africa is often framed as the world's poorest continent, yet it is also incredibly rich—in linguistic, ecological, and resource diversity. Initially, we even considered highlighting musical and ecological wealth to challenge conventional ideas of value. Ultimately, we focused on natural resources, not arbitrarily, but to emphasize how what is perceived as wealth can in fact be a curse—the so-called "resource curse." When people are denied control over their land and wealth, those assets can become liabilities. ***Our map aims to draw connections between resource extraction and poverty, prompting viewers to ask: what is poverty, who defines it, and who benefits from these definitions and systems?*** By contrasting traditional measures with deeper questions about power and ownership, we hope to challenge dominant narratives and foster a more critical understanding of poverty.

WHAT'S BENEATH THE POVERTY LINE?

This map highlights a striking paradox: many African countries remain below the extreme poverty line of $2.15/day, as defined by the UN and World Bank, yet they are among the most richly endowed with natural resources on Earth. From cobalt in the Congo to oil in Angola and gas in Algeria, these resources fuel the global economy — but rarely benefit the majority of people living where they're extracted.

This disconnect is no accident. It points to deeper structural issues in the global system of production and consumption — systems that prioritize profit over people and perpetuate extractive relationships between the Global North and South. Achieving Sustainable Development Goal 1: No Poverty by 2030 will not be possible through surface-level solutions alone. It requires a fundamental rethinking of who controls resources, who profits from them, and what "development" really means.

Africa is often seen through the lens of poverty. But perhaps we should ask: what is poverty? Is it merely income-based, or does it also reflect imposed narratives that obscure wealth in other forms — ecological, cultural, linguistic, communal? This map invites us to reconsider not only how we define poverty, but who benefits from keeping that definition in place.



2 ZERO HUNGER

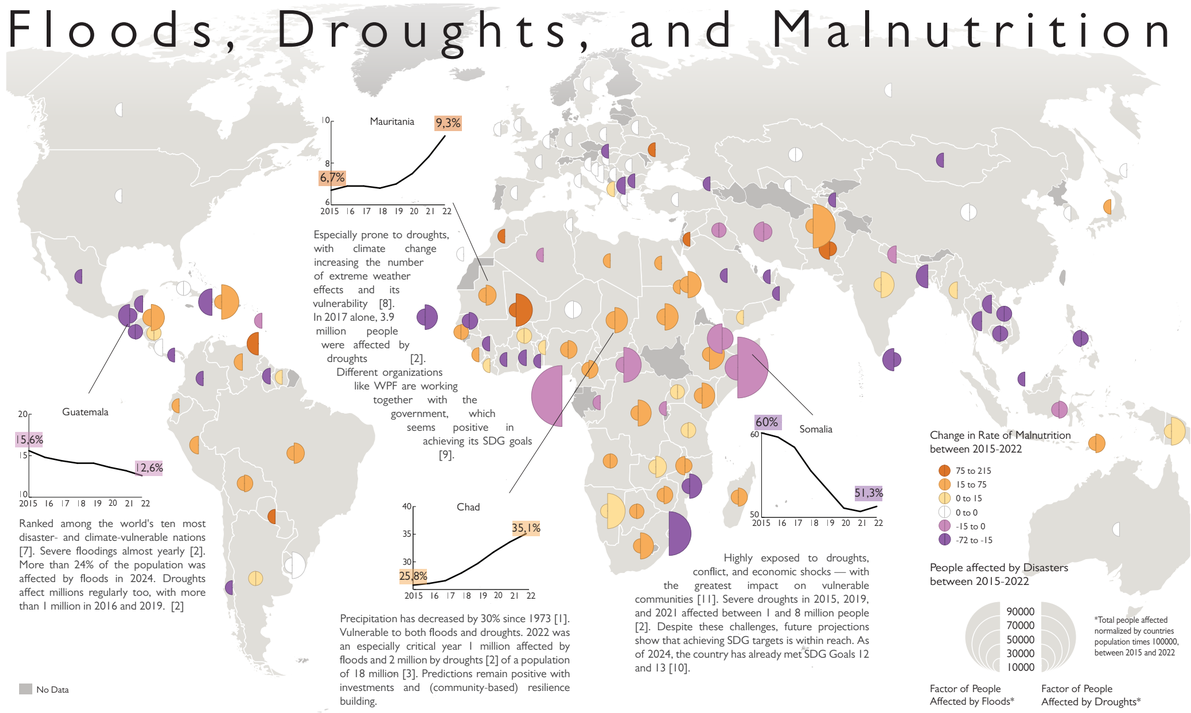
Floods, Droughts, and Malnutrition

Gabriela Spriestersbach & Lilly Tockner

At the start of the Design Challenge, we both initially selected a different SDG, but later shifted to ***Sustainable Development Goal 2 on Zero Hunger,*** as our original choice already had team representation. We pivoted to Zero Hunger because of its critical importance in promoting global food security and its strong links to other global issues such as poverty, climate change, and inequality. During the early brainstorming and mind mapping process, we began identifying key factors that disrupt the food supply chain, such as restricted access to conflict zones. Our focus then expanded to the impact of climate-related disasters on food systems. Through research, we discovered the EM-DAT database, which provides comprehensive global data on natural disasters. We specifically collected data on droughts and floods from 2000 to 2024. We also collected the data on malnutrition rates and the Food Price Anomalies Index from the official SDG database. Our team began organizing these datasets to uncover potential correlations between climate events and food insecurity.

During the event, our design process began with exploring various indicators related to SDG 2, focusing particularly on the causes and impacts of malnourishment. We were initially drawn to the role of food price anomalies and started examining their relationship with natural disasters alongside the rate of change in malnutrition prevalence. We arbitrarily chose to focus on floods at first and considered creating two separate maps, one for disaster data and another for malnutrition. However, after receiving feedback, we refined our approach and decided to integrate the data into a single map using proportional symbols. We used the size of the symbols to represent the number of affected people and color to indicate whether malnutrition had increased or decreased. At this stage, we also included droughts in our analysis and represented both disasters by splitting the proportional symbol accordingly. One of our early challenges was narrowing down our ideas, as there were many possibilities and some were too ambitious for the scope of the Design Challenge. As we progressed, we gradually scaled back our work to maintain a clear and concise narrative. This helped us stay focused and ensured our final design effectively communicated our insights.

Our final design explores the complex relationship between climate-related disasters and malnutrition, focusing on SDG 2 on Zero Hunger. Using data from 2015 to 2022, our map visualizes the number of people affected by floods and droughts worldwide and compares this with changes in the prevalence of malnutrition. We selected four countries with contrasting profiles—one country highly affected only by droughts (Mauritania), one primarily by floods (Guatemala), two with both, but either an increasing (Chad) or decreasing malnutrition rate (Somalia)—to highlight that the relationship is not straightforward. ***Many factors play a role in the prevalence of food insecurity and malnutrition too, including conflict, displacement and the economy. With our map we want to raise questions on the influences and impacts of disaster on malnutrition.***



Map Data Sources: United Nations Statistics Division. (n.d.). SDG Indicator 2.1.1: Prevalence of undernourishment [Database]. UN SDG Data Portal. https://unstats.un.org/sdgs/dataportal/database; Centre for Research on the Epidemiology of Disasters (CRED). (n.d.). EM-DAT: The international disaster database. https://www.emdat.be/; UN Geospatial Hub. UN Geodata simplified in GeoJSON https://geoportal.un.org/arcgis/apps/sites/#/geohub/datasets/d7caaff3ef4b4f7c82689b7c4694cd92/about.
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3 GOOD HEALTH AND WELL-BEING

How Many Health Workers Does it Take to Meet People's Real Health Needs?

Betty Selena Castro Benavides & Beatriz Oliveira de Carvalho

Our team chose to focus on ***Sustainable Development Goal 3 on Good Health and Well-Being***, recognizing that health is a fundamental human right and essential for people to fully access their other rights. We selected this SDG as our top choice based on our previous experiences working on health-related issues in our home countries. Beatriz has researched the relationship between safety perception and walkability, while Betty contributed to child malnutrition programs at the Ministry of Public Health. Initially, we were interested in exploring topics directly related to population health, such as substance abuse or communicable diseases. However, as we reflected on systemic barriers, we realized the importance of addressing one of the foundations of any health system: the health workforce.

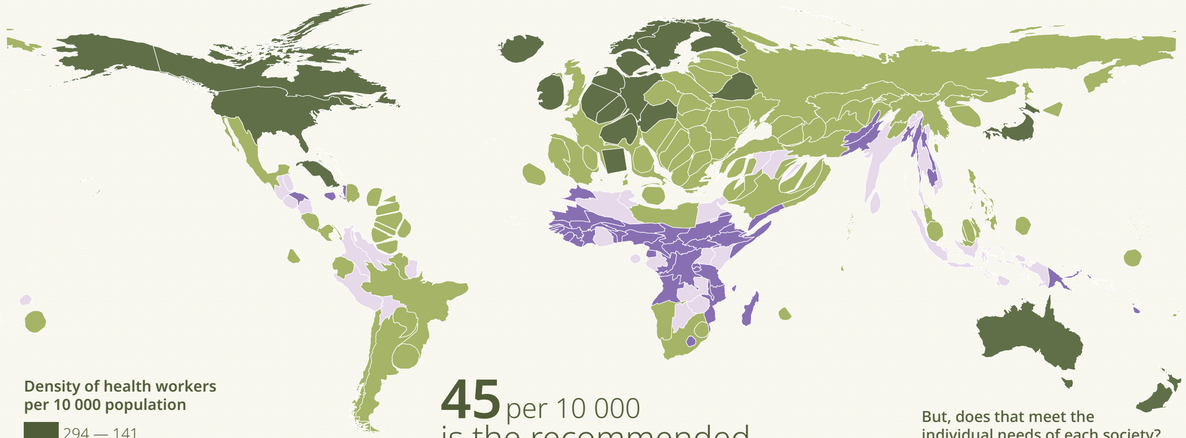
Our design process began with an exploratory analysis of the UN Tier I and II health dataset. Early on, we realized that many countries do not consistently report their indicators, making it difficult to form a comprehensive global overview. To address this, we used the World Health Organization (WHO) dataset, which provided more up-to-date and detailed country-level data. With this, we defined two main goals: first, to highlight inequalities in the distribution of health workers (doctors, nurses, and midwives) across countries, and second, to identify a threshold that indicates whether a country has enough health workers. The 2016 WHO's "SDG index threshold" of 4.45 health workers per 1,000 people helped us establish this benchmark. To communicate our findings, we used a cartogram. The area distortions helped emphasize disparities between countries. We also applied a diverging color palette centered around the threshold value to clearly differentiate countries above and below this minimum density. Through this process, we learned the importance of selecting the right data sources and visual techniques to clarify complex global inequalities.

Globally, the number of health workers may be enough to meet the minimum threshold needed to serve the world's population, but this workforce is not equally distributed. By using the WHO's benchmark of 4.45 doctors, nurses, and midwives per 1,000 people, we aimed to provide a global reference point for understanding which countries might be meeting basic health workforce needs. However, this threshold should not be interpreted in isolation. Population characteristics, local health priorities, and regional health system structures play a key role in determining how many health workers are truly needed in a given context. A number that is sufficient in one country may be inadequate in another. ***Health is not only a technical issue—it is deeply social and cultural. Systems must adapt to the specific needs, values, and contexts of each society. O***ur design emphasizes that while global indicators are helpful for comparison, achieving equitable health for all requires localized understanding and targeted investment in the health workforce where it is most needed.

How many health workers does it take to meet people's real health needs?

68 per 10 000 is the current **global mean**
— but thats not equally distributed

This map shows the **density of physicians (medical doctors), nurses, and midwives**, with each **country's area distorted** according to that density — revealing deep inequalities in global access to health care.



Density of health workers per 10 000 population

- 294 — 141
- 140 — **45**
- 44 — 22
- 21— 3

Source: Global Health Workforce statistics database (WHO, 2025)
Org.: Beatriz Oliveira de Carvalho & Betty Selena Castro Benavides (2025)

45 per 10 000 is the recommended **minimum threshold**
— as defined by the SDG index threshold (2016)

This means that if health workers were equally distributed, the current global workforce **would already be enough to guarantee health to all**.

But, does that meet the individual needs of each society?

While this threshold offers an important insight for the global health workforce, **population characteristics and local health** system priorities can **shift the minimum** required density. It's also important to remember that national numbers can hide internal disparities.

Better Students, But Not Worse Employees? Education and the Gender Pay Gap

Agata Kiseleva & Emiel Verté

Our team chose ***Sustainable Development Goal 4 on Quality Education*** because we believe that education is a foundational right and a key factor in reducing inequality and promoting sustainable development. At the start of the Design Challenge, we were particularly interested in understanding disparities in educational outcomes across gender. We focused on Indicator 4.1.1, which measures the proportion of students in lower secondary education who reach at least minimum proficiency in reading and mathematics. Initially, we intended to analyze global data, but we quickly encountered a significant obstacle: many countries lacked sufficient or consistent data. As a result, we narrowed our scope to focus on Europe, where the data was more complete and comparable. To enrich our analysis and highlight the persistent link between educational and economic inequalities, we decided to include a second variable: the unadjusted gender pay gap.

During the design process, our initial idea was to create a proportional symbol map where each symbol would be split into two parts: one representing the percentage of boys achieving minimum proficiency in education and the other representing the percentage of girls. However, this approach proved visually ineffective. The differences between the two values were often small, and when scaled to fit the map, these differences became nearly invisible, making the symbols difficult to interpret and the comparison unclear. This challenge forced us to reconsider our visual strategy. After testing different design alternatives, we decided to calculate the education gender gap as a single value (female proficiency minus male proficiency) and represent it through color intensity. This made the map much clearer and allowed us to easily communicate which gender was performing better and by how much.

To add more depth and draw attention to broader gender inequality, we introduced a second variable: the gender pay gap, represented by symbol size. Combining these two indicators allowed us to visualize both educational outcomes and economic disparities in a single, meaningful layout. Along the way, we learned the importance of simplicity in map design and how data visualization must adapt to the nature of the data.

Our final map tells a layered story about gender inequality in education and the workforce within the context of SDG 4 on Quality Education. Using the latest available data from 2018 to 2023, we focused on Indicator 4.1.1, which measures the proportion of lower secondary students achieving minimum proficiency in reading and mathematics. We visualized the gender education gap by subtracting male proficiency from female proficiency, so positive values highlight countries where girls outperform boys. We discovered that in many European countries, girls outperform boys in educational achievement. However, when we added the unadjusted gender pay gap as a second variable, represented by symbol size, a striking contrast emerged: even in countries where girls perform better in school, women still earn significantly less than men on average. This revealed a disconnect between educational success and economic outcomes, emphasizing that access to and success in education alone do not guarantee gender equality in the labor market. ***Our take-home message is that while SDG 4 focuses on ensuring inclusive and equitable education, it must be understood in relation to broader systemic issues.*** Bridging the gap between educational achievement and real-world gender equality remains a key challenge on the path to sustainable development.

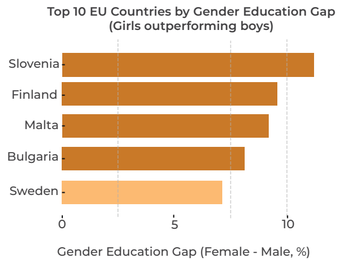
Better Students, but not Worse Employees?

Education and the Gender Pay Gap

In European countries, on average, **5.5% more girls** reach the minimum literacy goals than boys, with some countries showing a gap of up to 10.66%

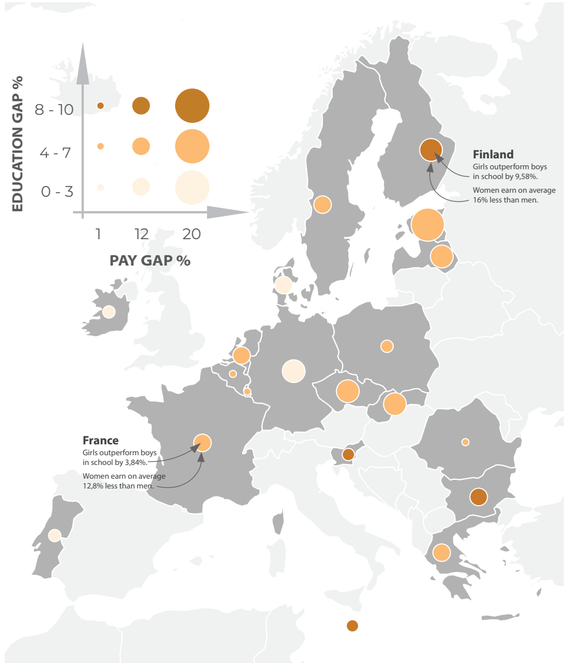
Despite this, **women in the EU earn, on average, 12% less per hour** than men.

This map uses Indicator 4.1.1 (latest data from 2018–2023), which measures the proportion of young students in lower secondary education who achieve at least minimum proficiency in reading and mathematics.



The gender education gap is calculated by subtracting the proportion of males reaching this level from the proportion of females. A positive value means girls outperform boys.

The color intensity represents the size of this performance gap—the darker the color, the larger the difference in favor of girls. The symbol size represents the unadjusted gender pay gap: the difference in average gross hourly earnings between women and men. A larger symbol indicates a larger pay gap



Authored by Agata Kiseleva, Emiel Verté
Data: Eurostat (2025), United Nations (2025)

5 GENDER EQUALITY

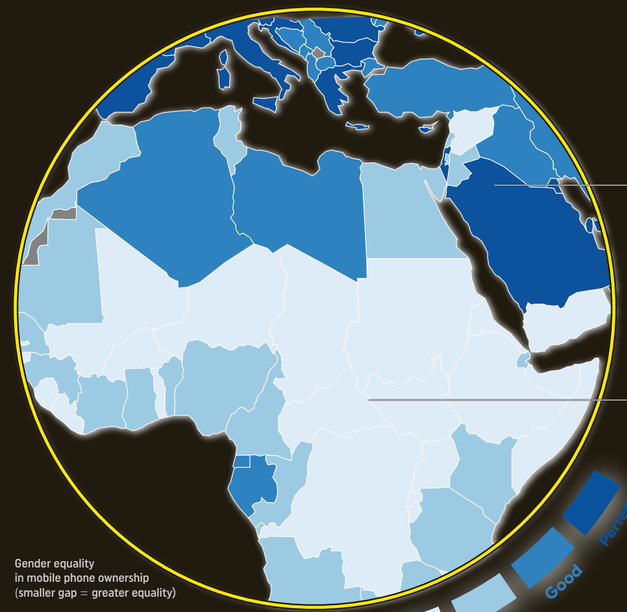
Gender Inequality in Mobile Phones

Ximing Wang & Yvonne Darko

Our team chose ***Sustainable Development Goal 5 on Gender Equality*** because we believe equal opportunities for women make a big difference. At the start of the Design Challenge, we examined global mobile-phone ownership data and found that in some regions fewer than 50 percent of women have a phone compared with almost 90 percent in others. We saw that not having a phone keeps women from accessing news, jobs, and services. So, we focused on the digital divide for women. We want to understand barriers like device cost in lower income areas, low digital skills, and social restrictions. We hope that through this map, people will become aware of this situation and take gender inequality in mobile phone ownership seriously.

During the Design Challenge, we wanted a map that clearly showed the gap between men's and women's phone ownership. At first, we thought pie charts would do the job, but the numbers did not line up well and the pies looked messy on a map. So we switched to a color-shaded map with a zoomed-in box for North and Central Africa. We cleaned and organized our data in ArcGIS Pro, then moved the shapes and tables into Adobe Illustrator to polish the look. Highlighting African regions shows the biggest gaps without crowding the world map. Our first idea with chart overlays ended up distracting from what mattered. Along the way, we learned to keep things simple: use only a few colors, make sure the most important parts stand out, sketch lots of drafts, get feedback, and tweak sizing and colors until the design is balanced.

Our map tells a straightforward story: women's phone ownership is not the same everywhere. We use dark blue for places where women and men have phones at nearly the same rate, and pale blue where women fall far behind. By zooming into North and Central Africa—the lightest spots with the lowest rates—you can see countries where fewer than half of women own a phone. This gap is not just a number: a basic handset can cost up to 25 percent of a woman's income, training is hard to come by, and social rules often keep women offline. Without a phone, women miss out on news, work opportunities, and a chance to join community conversations. ***Our take-home message is clear: fixing this digital divide cannot wait. We need cheaper devices, hands-on training in local communities, and policies that support women's access to technology.*** Only then can we close the gender gap and make real progress and beyond.



What drives the gender gap in mobile phone ownership?

Lowest Gender Gap Area

96.1% Proportion of women who own mobile phones

93.2% Proportion of women who use the Internet

Highest Gender Gap Area

47.2% Proportion of women who own mobile phones

20.9% Proportion of women who use the Internet

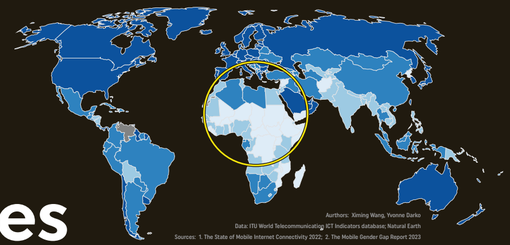
Cultural Norms
In Pakistan fewer women are reported to use mobile phones because of the lack of approval from their families.[2]

Digital Literacy
In Bangladesh, India and Guatemala, fewer women use the internet because they cannot read and write.[2]

Economic Barriers
In Low and middle income countries an entry level phone takes up 25% of a woman's income.[1]

Perceived Impact
In Ethiopia 20% of female respondents and 10% of male respondents perceive the Internet has a negative impact.[2]

Gender Inequality in Mobile Phones



Authors: Ximing Wang, Yvonne Darko
Data: ITU World Telecommunication/ICT Indicators database; Natural Earth
Sources: 1. The State of Mobile Internet Connectivity 2022; 2. The Mobile Gender Gap Report 2023

6 CLEAN WATER AND SANITATION

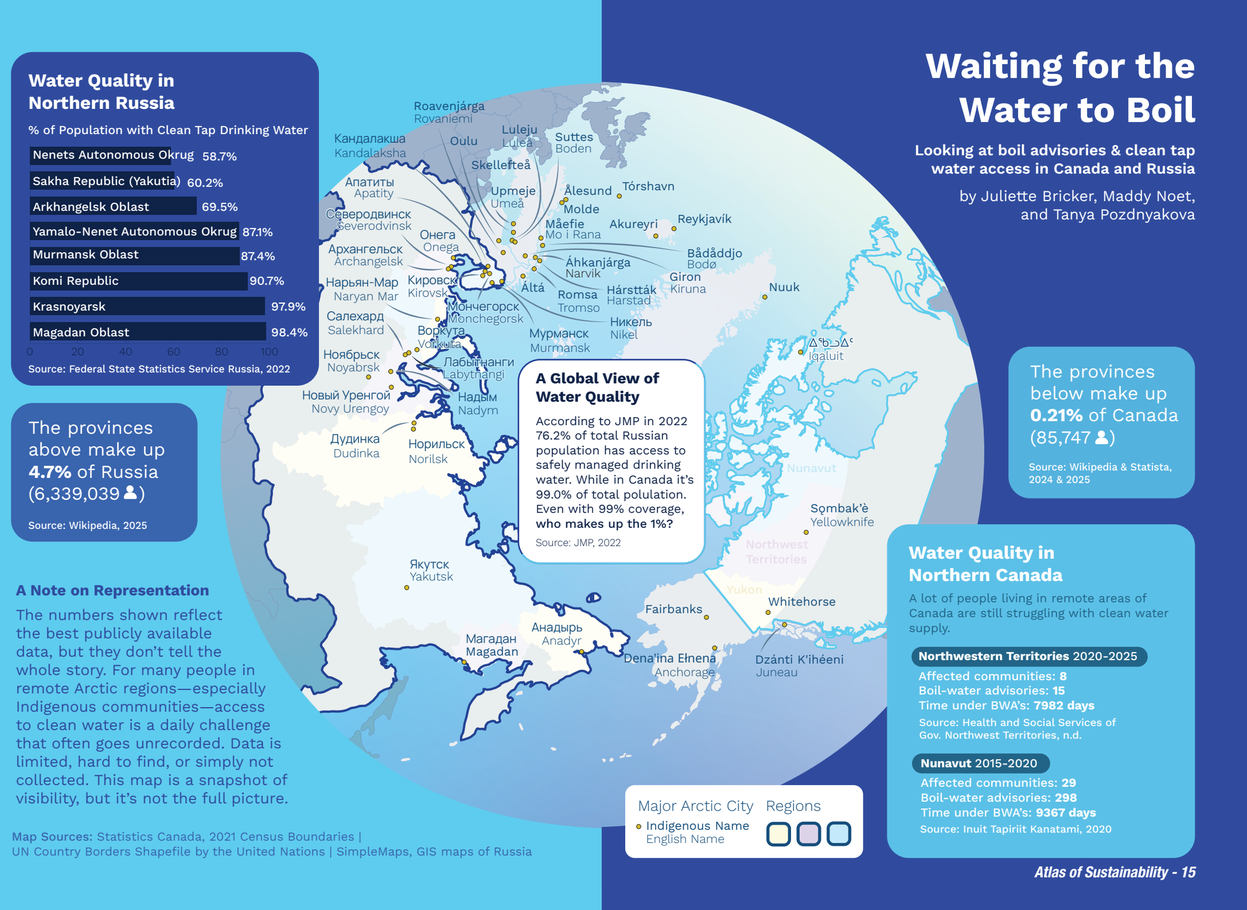
Waiting for the Water to Boil

Juliette Bricker, Maddy Noet, & Tanya Pozdnyakova

We selected ***Sustainable Development Goal 6 on Clean Water and Sanitation*** because we thought it would be interesting to research something that we depend on constantly in our daily lives. After a little bit of digging, we found lots of readily available data on those in heavily populated urban areas throughout the U.S. and Europe. However, we decided that we wanted to represent and research those living in more remote and rural areas. We challenged ourselves to think of places outside of hot climates that may struggle with consistent water access. After a bit of searching, we decided we could tell a powerful story about people in the Arctic and the unique challenges that come with their water infrastructure, especially given the harsh environmental conditions and limited resources.

This project was defined by our effort to find recent data from the Arctic that was relevant and usable to the subject matter. We could see from several news reports that there is an issue with accessing clean water in rural areas throughout the Arctic Regions, but could not find any datasets that represented this problem. We tirelessly looked for data to use, and this became the overarching story of our map. We learned the importance of data and a lot about the water access issues going on in the Arctic. After collecting our data we worked in Figma as it is very efficient for sorting colors and labels. It is also collaborative, unlike Adobe Illustrator. We made many sketches, then drafted in black and white before making our many iterations in color. In the end, we transferred the content to Adobe Illustrator. We learned to work under time pressure and display data, or a lack thereof for the story we wanted to tell.

Our final design focuses on SDG 6 on Clean Water and Sanitation, specifically the inequities in access to clean tap water in remote northern regions of Canada and Russia. While both countries boast high national averages for access to safely managed drinking water, these figures mask deep regional disparities—especially in Arctic and Indigenous communities. Our map tells the story of those left behind: populations living under long-term boil water advisories or with limited access to clean water infrastructure despite residing in wealthy, developed nations. The data reveals that access to safe drinking water is not just a developing world issue, but a critical, ongoing challenge in high-income countries as well. Our design highlights not only statistics, but the human cost of policy neglect and infrastructural failure in the most remote and vulnerable areas. ***The key takeaway is that SDG 6 cannot be measured only by national averages—it must account for those in isolated regions whose experiences often go undocumented.*** Clean water is a human right, and this design advocates for more inclusive data collection, community-driven solutions, and equitable resource distribution to ensure all communities are counted—and served—in the push for sustainable development.



7 AFFORDABLE AND CLEAN ENERGY

From Fossil Fuels to Renewables: Is the World Making the Switch?

Matteo Duniš & Suphaphong (Tobias) Sirinapaphan

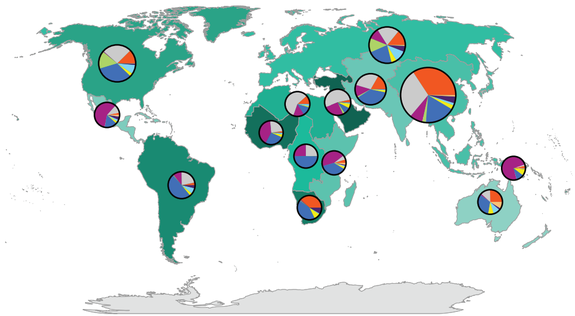
Our team chose ***Sustainable Development Goal 7 on Affordable and Clean Energy***, which aims to provide everyone with access to affordable, reliable, and sustainable energy. We were drawn to SDG 7 because of its strong connections to climate change, sustainable development, economic opportunity, and global equity. From the beginning of the Design Challenge, our goal was to visualize the global progress of increasing the share of renewable energy in electricity generation. Additionally, we aimed to showcase electricity production across different regions and highlight the most widely used energy sources. Ultimately, we created two world maps. The first map shows how electricity generation is divided by source in different regions, enabling users to compare the dominance of fossil fuels, renewables, and other sources. The second map highlights countries that have made significant progress or experienced setbacks on their path toward renewable energy adoption. By focusing on electricity production rather than consumption, we aimed to emphasize the importance of clean energy transitions at the system level and draw attention to infrastructures in need of transformation.

At the beginning of Design Challenge, our plan was to create five maps: two world maps and three focused on countries with the largest shares of renewable energy. However, given the time constraints and complexity of the data, this proved too ambitious. Ultimately, we decided to focus on two world maps and two slopegraphs. The biggest challenge throughout the process was visualizing the data in an accurate and easy-to-understand way. Since both maps included a large amount of information, we struggled to decide on the most effective way to display it. For the first map, the main issue was representing the different energy sources while allowing for meaningful regional comparisons. After experimenting with several techniques such as doughnut charts and polar bar charts, we chose proportional pie charts; however, balancing visual clarity and detail remained difficult. For the second map, our goal was to show how countries progressed or regressed over time in terms of their renewable energy share. Making this change visible and easy to interpret was challenging, especially when trying to avoid overwhelming the viewer. Through these challenges, we learned the importance of simplifying complex information and making deliberate design choices to ensure clarity and impact.

Our final design tells a story about the global transition toward renewable energy, highlighting both progress and persistent disparities in the pursuit of SDG 7. The pressing need to transition from fossil fuels to cleaner, more sustainable sources of electricity is crucial in the face of climate change and to guarantee energy access for all. Our first world map illustrates the distribution of electricity generation by energy source across different regions, revealing striking contrasts. For instance, it shows regions that are heavily dependent on fossil fuels versus regions that have significant shares of hydro, wind, or solar power. This map emphasizes the uneven starting points and capacities that countries have in making the energy transition. The second map, which is supported by slopegraphs, tracks changes in the renewable energy share over time. It shows which countries have made significant strides and which have fallen behind. We illustrate the key challenge that, while global renewable energy capacity is growing, progress is far from equal, and some countries are regressing. ***The main takeaway is that the transition to clean energy requires not only technological solutions, but also policy changes, investments, and equitable practices.*** To achieve SDG 7, more coordinated and inclusive efforts are needed to support lagging countries.

From Fossil Fuels to Renewables: Is the World Making the Switch?

Tracking Energy Diversity Across Subregions in 2022



Sustainable Development Goal 7 (SDG 7) aims to ensure access to affordable, reliable, sustainable, and modern energy for all by 2030. Energy is fundamental to addressing major global challenges, from climate change and economic growth to health and education. The goal focuses on expanding electricity access, increasing the share of renewable energy, and enhancing energy efficiency. Reaching these targets is vital for reducing emissions and achieving equitable development, particularly in regions still facing energy poverty.
Countries differ greatly in how they produce electricity, shaped by their natural resources, infrastructure, and policy choices. Some benefit from hydropower due to abundant water systems, while others harness solar, wind, or geothermal energy depending on their geographic conditions. Yet, fossil fuels—such as coal, oil, and gas—still account for a significant share of electricity. The pace of transitioning to renewables varies widely, influenced by economic capacity, political will, and access to technology. This uneven shift underscores the importance of tailored energy strategies that address local needs while advancing global sustainability goals.

Electricity Production by Source (2022)

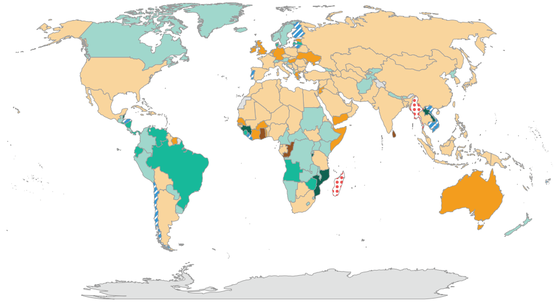
Coal, Gas, Oil, Nuclear, Hydro, Solar, Wind, Bioenergy, Other

Electricity Production in TWh (2022)

10 000
5 400
320

Since the adoption of the Sustainable Development Goals in 2015, the global community has intensified its focus on transitioning to renewable energy sources. This thematic map illustrates the shift in renewable electricity generation share by country between 2015 and 2022, highlighting where progress has been made and where setbacks have occurred. By visualizing these changes, we aim to assess how countries are aligning with the SDG target of increasing the share of renewables in the global energy mix. The map reveals diverse trajectories: some countries have significantly expanded their renewable capacity, marking notable progress, while others show stagnation or regression, underscoring the uneven pace of the energy transition across the globe.

Who's Moving *Forward* *Backward* in Renewable Electricity Generation?



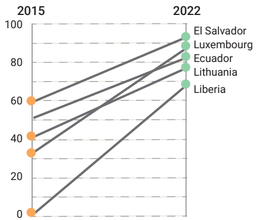
X: Share of RE Generation

>50% <50%

+10%
-10%

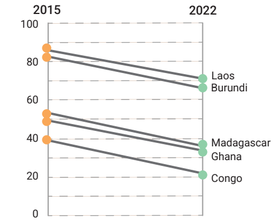
Y: RE Share Shift (2015-2022)

Top 5 Countries Advancing in Renewable Electricity Transition (2015–2022)



Green indicates countries above 50% in both years, and orange shows those below 50% throughout. Shade intensity reflects change: bright for increases over 10 percentage points, light for minimal change (±10%), and dark for decreases over 10%. Overlay patterns mark threshold shifts: a blue horizontal line for progress (below to above 50%) and a red dot pattern for regression (above to below 50%).

Top 5 Countries with the Largest Decrease in Renewable Electricity Share (2015–2022)



Authors: Suphaphong (Tobias) Sirinapaphan, Matteo Dunis
Data Source: Ember (2024); Energy Institute - Statistical Review of World Energy (2024)

8 DECENT WORK AND ECONOMIC GROWTH

Moving Towards Free Labor: Stories of Fights for Government Compliance

Madeleine Boyles & Viera Muziwakhe Mlilo

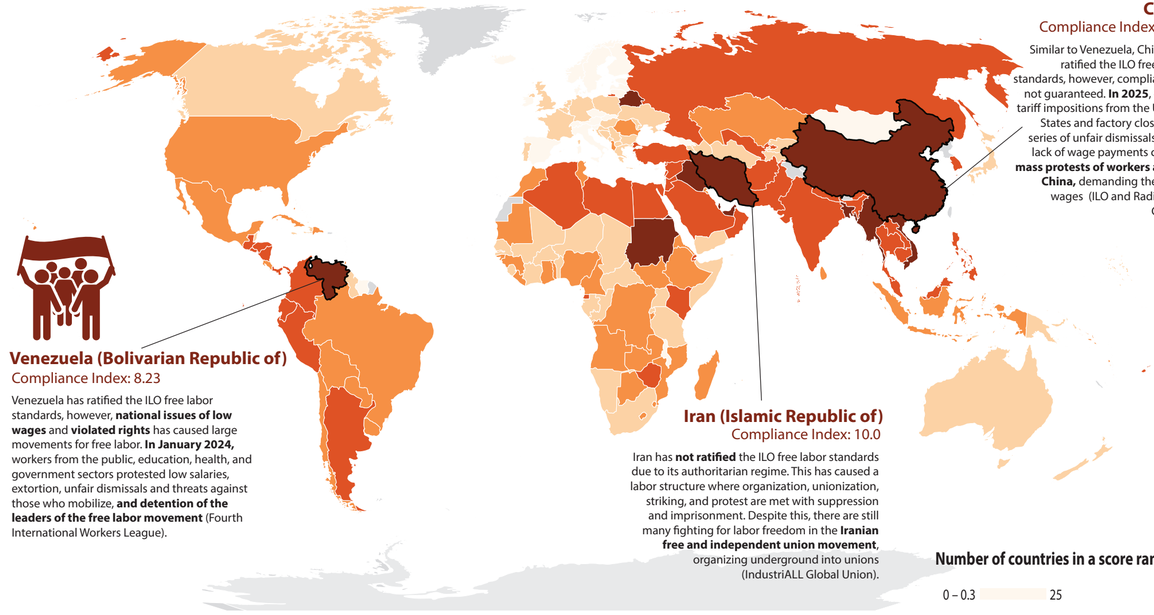
We selected ***Sustainable Development Goal 8 on Decent Work and Economic Growth***, focusing on free labor (i.e., freedom of association and collective bargaining) and government compliance with international conventions. Our target indicator was 8.8.2, which tracks the level of compliance among countries to fundamental labor rights following a standard defined by the International Labor Organization (ILO). Initially, we wanted to see how migrants and women fare within these dynamics among the different nations. However, the available SDG dataset did not have a detailed breakdown by these demographics. Despite this limitation, we decided to continue to make a visualization out of labor rights because we felt it was still a story worth telling. We felt like access to decent work is very important because it is one of the most defining features of a healthy society.

We started with a rough paper sketch to brainstorm how the different elements on the map could be placed. Then we used Indicator 8.2.2 to make a map, which was initially a basic choropleth map visualizing the data in equal intervals. From there, we adjusted the classification system to fit the positive skew of the dataset, which better balanced the geometric pattern of the data distribution. After that, we started testing different color palettes to see which ones better evoked a congruent emotion with how we wanted the map reader to intuitively approach the map. This included adding "highlights" to the countries where we had stories. It was quite tricky, but we got to a point where we felt we had the right balance of concern with hope. We then introduced text elements like the title and visual elements like the legend for context. We worked on balancing these newly added elements with the existing map using the grid and guides to maintain aharmonized layout. Our initial paper sketch did not look as good on the screen as we had hoped, so we had to move around the map elements to get the appropriate look and design. Ultimately, we are very happy with our design and its visual balance.

Our final design discusses "decent work" in terms of free labor by telling current stories from three countries where workers are protesting the lack of national compliance with the International Labor Organization's freedom of association and collective bargaining standards. We both thought that the ILO index, specifically on free labor, did not focus enough on the human experience of suppression, low wages, and attacks on unionization. As such, we decided to focus on countries where government compliance is low and the fight for free labor rights is a current struggle. We wanted to make a point about the specific challenges workers face due to low national compliance, and the steps they are taking to overcome these challenges. While SDG 8 mostly includes indicators related to gross domestic product (GDP), we wanted to focus on the human side of labor. ***The global economy is based on human labor and protecting the safety, security, and freedom of labor, in our opinion, was the most important dimension of this SDG to highlight.***

Moving Towards Free Labor: Stories of Fights for Government Compliance

An index ranking **national compliance** with the International Labor Organization's (ILO) free labor standards on **freedom of association** and **collective bargaining**, from 0 (complies fully) to 10 (no compliance).



China
Compliance Index: 9.13

Similar to Venezuela, China has ratified the ILO free labor standards, however, compliance is not guaranteed. **In 2025**, due to tariff impositions from the United States and factory closures, a series of unfair dismissals and a lack of wage payments caused **mass protests of workers across China,** demanding their due wages (ILO and Radio Free China).

Venezuela (Bolivarian Republic of)
Compliance Index: 8.23

Venezuela has ratified the ILO free labor standards, however, **national issues of low wages** and **violated rights** has caused large movements for free labor. **In January 2024,** workers from the public, education, health, and government sectors protested low salaries, extortion, unfair dismissals and threats against those who mobilize, **and detention of the leaders of the free labor movement** (Fourth International Workers League).

Iran (Islamic Republic of)
Compliance Index: 10.0

Iran has **not ratified** the ILO free labor standards due to its authoritarian regime. This has caused a labor structure where organization, unionization, striking, and protest are met with suppression and imprisonment. Despite this, there are still many fighting for labor freedom in the **Iranian free and independent union movement**, organizing underground into unions (IndustriALL Global Union).

No Data 0 0.3 1.6 3.3 7.0 10

Data from the United Nations Sustainable Development Goal Data Portal

The ILO was founded in 1919, and has advocated for workers' rights ever since. This index focuses on a nation's compliance with the ILO's more specific standards of free labor, like unionization, collective bargaining, striking, establishment and joining of organizations, and civil liberties.

Number of countries in a score range

0 – 0.3	25
0.3 – 1.6	71
1.6 – 3.3	46
3.3 – 7.0	37
7.0 – 10	11

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Bridging the Final 5%: Gaps and Opportunities in Global Connectivity

Weiyi Gao & Lingyu Tang

Our team selected ***Sustainable Development Goal 9 on: Industry, Innovation and Infrastructure*** because we believe that sustainable and inclusive development cannot occur without accessible infrastructure and equitable digital connectivity. After reading the 2024 SDG Report, we were particularly interested in the final 5% mobile broadband coverage gap, because it is a critical gateway to the internet in many developing regions. The fact that around 5% of the world's population still lacks mobile broadband coverage, with the majority residing in the most vulnerable areas such as Sub-Saharan Africa and remote island nations, raised important questions for us about equity, development, and global priorities. We wanted to explore how digital exclusion relates to infrastructure needs and whether current patterns of international aid align with these connectivity gaps. Our goal is to create a data-driven storytelling map that not only visualizes where people remain unconnected but also links those areas to international infrastructure support, revealing both missed opportunities and areas for targeted intervention. This approach would allow us to visualize data, spark discussion about digital inclusion, and emphasize that closing the "final 5%" coverage gap is essential to achieving the broader ambitions of SDG 9.

We began by identifying the most relevant indicators on our map theme: Proportion of population covered by at least a 3G mobile network and total official flows for infrastructure by recipient countries, which helped shape the narrative focus of our map. Then we retrieved data from the SDG global indicator database provided by the United Nations. To make our story coherent and the data more digestible, we decided to break the story into three clear visual components: a line chart showing 3G coverage trends, a global map of mobile broadband accessibility, and a thematic map of infrastructure aid in Africa. However, we faced challenges aligning spatial data layers, particularly when attempting to overlay international aid statistics with broadband gaps at different geographic scales. One major obstacle was the overlaps would lead to confusion visually. Accordingly, we decided to visualize the two indicators separately by two maps. We experimented with hexbin maps to visualize official flows for infrastructure, and struggled with placing them meaningfully. Throughout the process, we learned the importance of data storytelling clarity, especially in balancing aesthetics with accuracy. Also, we realized that spatial storytelling is not just about maps—it is about crafting a narrative that connects people to the data and inspires action.

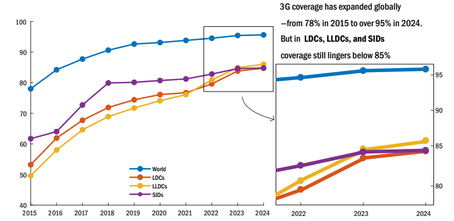
Our final design tells the story of the persistent mobile broadband coverage gap, focusing on the critical challenge of closing the final 5% gap. While 95% of the global population is now covered by 3G or higher mobile networks, the remaining 5% are concentrated in some of the world's most underserved and remote regions, particularly in Sub-Saharan Africa, Oceania, and Least Developed Countries (LDCs). These populations are not just disconnected from the internet—they are excluded from digital education, economic opportunity, and even basic services. Our story also explores whether international aid for infrastructure is adequately addressing this gap. ***By visualizing mobile broadband coverage alongside maps of official development assistance for basic infrastructure, we reveal a disconnect: some areas with low mobile broadband coverage receive very limited investment in digital infrastructure.*** To truly realize SDG 9, we need to ensure that future infrastructure investments prioritize inclusive digital access, especially in regions where it is still out of reach.

Bridging the Final 5%: Gaps and Opportunities in Global Connectivity

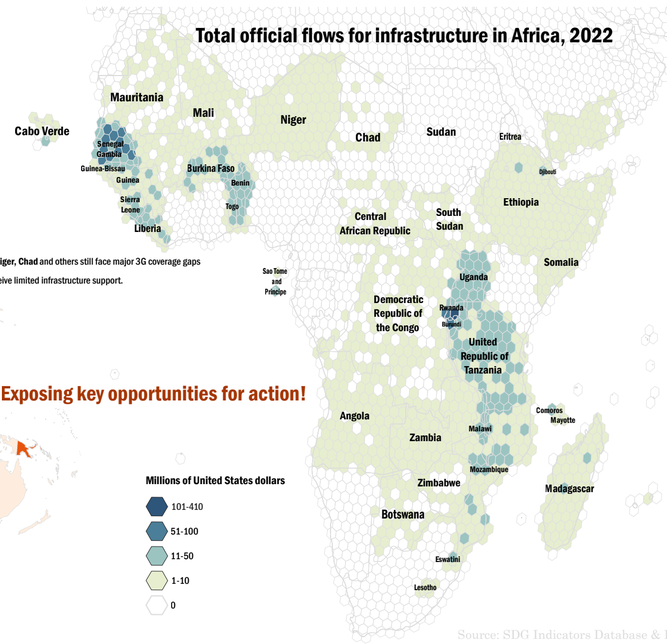
In our hyper-connected world, the Internet pulses like a heartbeat—linking people, powering economies, and opening doors to education and opportunity.
In many developing regions, mobile broadband—3G or higher—is not a luxury. It's a lifeline.

Proportion of population covered by at least a 3G mobile network (%), 2015-2024

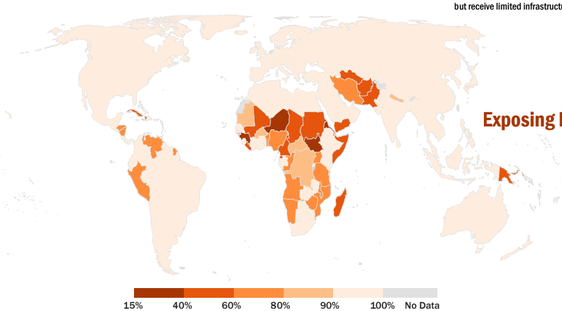
Since surpassing the 90 per cent mark in 2018, global 3G coverage has increased by only 4 percentage points.



Total official flows for infrastructure in Africa, 2022



Proportion of population covered by at least a 3G mobile network (%), 2022



Mali, **Niger**, **Chad** and others still face major 3G coverage gaps but receive limited infrastructure support.

Exposing key opportunities for action!

Source: SDG Indicators Database & ITU

10 REDUCED INEQUALITIES

Where is Life Better? Well-being Indicators in the EU and Turkey

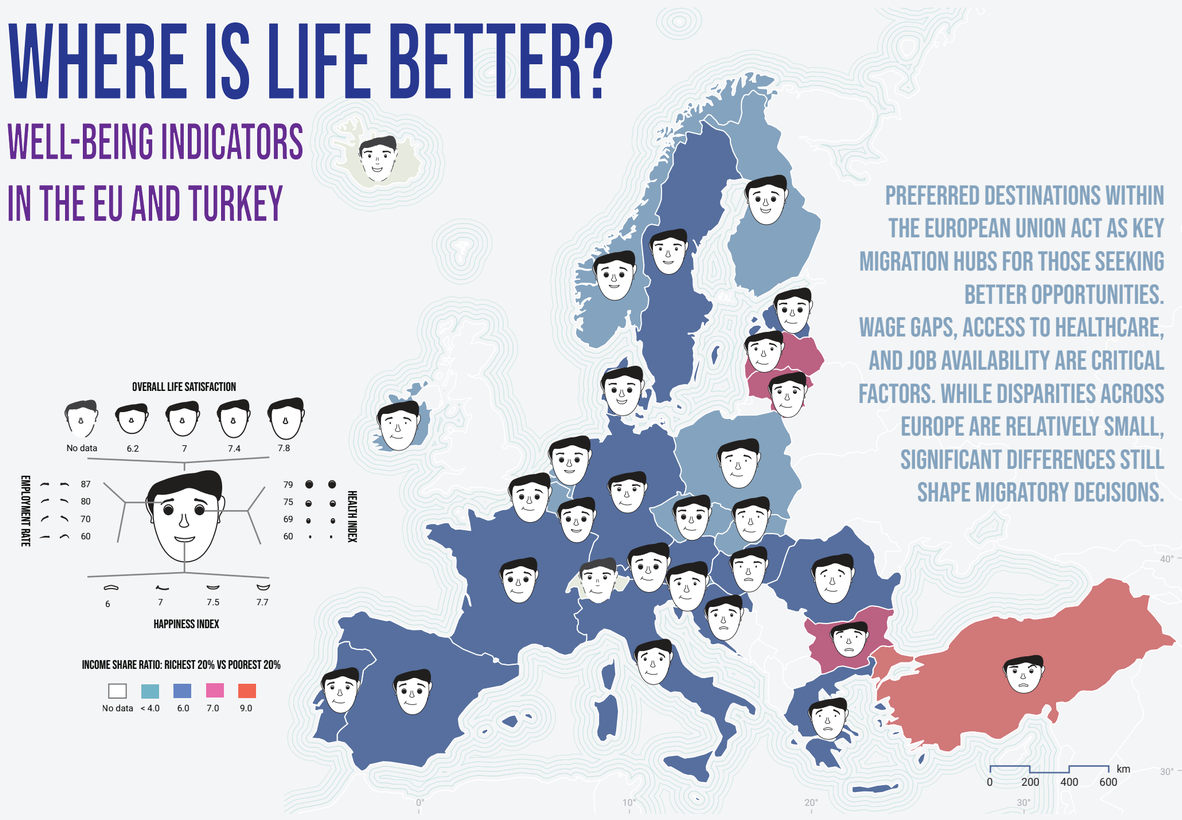
Kristina Dujakovic & Miguel A Martinez Solano

Sustainable Development Goal 10 on Reduce Inequalities aims to lower disparities within and among countries by narrowing income gaps, promoting inclusion, supporting vulnerable groups such as migrants, and facilitating safe and orderly migration. At the start of the Design Challenge, we chose to focus on Target 10.7, which emphasizes responsible migration and mobility. Our initial goal was to map migration flows caused by war, particularly the ongoing conflict between Ukraine and Russia. While we quickly found data on Ukrainian refugees in EU countries, comparable data on Russian migrants was scarce, and detailed refugee routes were unavailable. Due to these data limitations, we shifted our focus. Instead of mapping routes, we used available indicators—such as income inequality, employment rates, health index, happiness, and overall life satisfaction—to explore why certain countries are preferred destinations for migrants. These indicators also allowed us to show disparities not only between countries but within them. For example, high employment does not always align with a high happiness index. Our final map was inspired by the visual approach used in the Unclassed Chernoff Face Map from somethingaboutmaps.com, which helped us represent complex data through clear and engaging storytelling.

One of our main challenges was selecting a clear and specific topic for the map. Initially, we aimed to represent migration and forced displacement caused by war, using the conflict in Ukraine as a case study. However, due to limited and inconsistent data—particularly on migration routes and displaced populations—we decided to shift our focus. Instead, we explored the concept of quality of life across the European Union and used the Chernoff Faces method to visualize multiple indicators. It was our first time working with this technique, which made the process both fun and experimental. We spent time thinking through how to design an effective and readable layout. Throughout the project, we also improved our design skills using Adobe Illustrator and Inkscape. For Kristina, this was her first experience with design software, making the tools and techniques we learned especially valuable and empowering.

The main goal of this map is to illustrate the essential life conditions that many people seek in response to global inequality. Factors such as access to employment, quality healthcare, education, income security, and overall well-being are not equally distributed across the world, leading individuals to migrate in search of better opportunities. ***Within the European Union, economic disparities between countries—though relatively smaller than in other regions—still shape migration patterns. Countries with stronger economies, more robust welfare systems, and higher life satisfaction tend to become key destinations for both international and intra-European migrants.*** These movements occur not only for work or study but also for the pursuit of a higher quality of life.

The map aims to visually communicate these preferences and patterns by comparing indicators such as income inequality, health index, employment rate, and general life satisfaction. It also sheds light on the internal inequalities that persist within the EU, which continue to influence mobility across the continent. We chose to include Turkey in this analysis because, as an official EU candidate country, it actively contributes to EU statistical surveys and participates in broader regional policy discussions. Its inclusion provides a more comprehensive picture of migration dynamics and socio-economic contrasts in the region.



Urban Growth at a Cost: Population, Land, and Inequality in Cities

Lesly Aracelly Bautista Buendia & Artur Kislitsyn

Our team selected ***Sustainable Development Goal on Sustainable Cities and Communities***. Initially, our group was interested in focusing on disaster management, aiming to explore how communities respond to and recover from natural hazards. However, through further discussion and reflection, we realized that the impact of disasters is often intensified by the conditions in which people live. Inadequate housing, rapid population growth, and poorly planned urban expansion can make communities significantly more vulnerable to disasters. This insight led us to shift our focus toward issues such as population growth, land consumption, and inadequate housing. We recognized that many people are affected not only by the disasters themselves but also by the unsafe and unsustainable environments in which they reside.

Our group began the design process with a brainstorming session based on the indicators under SDG 11. We generated several ideas and, once we agreed on a concrete concept, we proceeded to download the specific information. However, as we explored the data more deeply, we realized that the format and scale of the data (whether city- or country-level) influenced the direction of our project. This required us to adapt our initial idea to better align with the available information. After visualizing the data, we focused on the design aspects of our presentation. Since the topic revolves around people's homes and living conditions—which can be a serious and sensitive issue—we intentionally chose to present it in a warm and inviting visual style. We wanted the final product to convey empathy and care, providing a comforting tone even though the subject matter is complex and often distressing. Through this process, we learned the importance of flexibility, the need to adapt to the realities of data, and how thoughtful design choices can shape the emotional impact of a message.

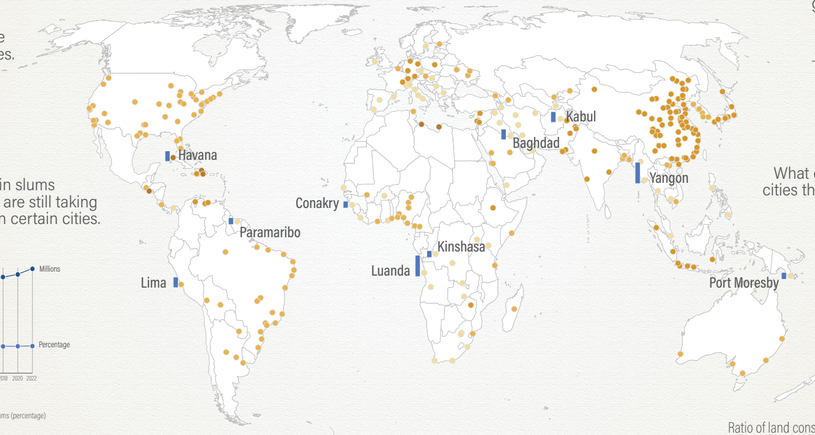
Cities around the world are expanding, with urban populations and the number of large metropolitan areas increasing at a fast pace. On the surface, data shows that the proportion of people living in slums is decreasing, but in reality, the total number of slum dwellers continues to rise. This means that although cities are developing, they are not always doing so in an inclusive or sustainable way. Our design highlights that in many cities, especially in developing regions, the growth of slums is particularly alarming. Moreover, cities are sprawling outward, taking up more land per person and often expanding into peri-urban areas where infrastructure and housing are inadequate. These trends raise serious concerns about land consumption, housing quality, and access to basic services. ***The key takeaway from our project is that sustainable urban growth is not just about managing population numbers—it is about ensuring that as cities grow, they do so in a way that provides safe, affordable, and adequate housing for all, without putting further pressure on land and resources.***

Urban Growth at a Cost

Population, Land, and Inequality in Cities

While the share of slums is decreasing in the world, there is a significant number of cities where substandard informal settlements are becoming more dominant. Urban sprawl and excessive land use also seems disturbing in the rapidly growing urban areas.

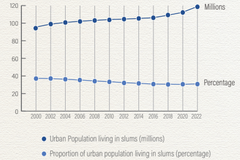
Urban population is growing rapidly.
From **51% (3.5B) in 2010**
to **57% (4.6B) in 2023,**
the most notable increase
is in the large metropolises.



There are cities where the proportional growth of slums is **alarming**

Proportional growth
of population in slums
from 2010 to 2020

Proportion of population in slums
is **decreasing,** but slums are still taking
up more and more area in certain cities.



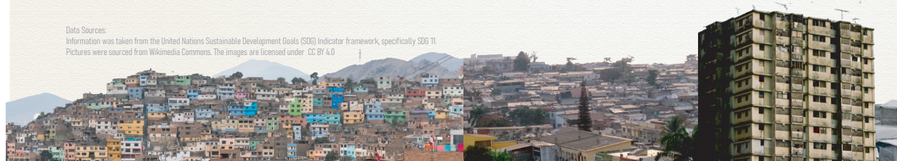
What causes the continued rise of
cities that are unfit for a decent life?

- Extensive urban planning
- Rapid urbanization
- Income disparity
- Suburbanization

Cities are growing
and **sprawling**, taking
more and more natural
area per citizen

Ratio of land consumption rate to population growth rate
(average for largest urban areas in the country)

Data Sources:
Information was taken from the United Nations Sustainable Development Goals (SDG) Indicator framework, specifically SDG 11.
Pictures were sourced from Wikimedia Commons. The images are licensed under CC BY 4.0



12 RESPONSIBLE CONSUMPTION AND PRODUCTION

A Closer Look at Plastic Waste in South East Asia

Paula Martín Castillo & Camille Rahier

Sustainable Development Goal 12 on Responsible Consumption and Production ensures sustainable food and material systems and services for nations, businesses, and individuals. This goal is a core part of the sustainable development of cities and society and aims to reduce the use of fossil fuels, natural resources, and food and material waste. The latter inspired us to explore plastic production and disposal, as we recognize it as a growing issue. Delving into the topic raised questions such as "How much plastic waste do we each produce?", "Where does it end up?", "How does this vary globally?", and "Who has to deal with it?". A pattern emerged: more developed nations produced very high amounts of plastic waste, while less developed nations produced very high amounts of plastic waste. However, the issue is much more complex than this. Plastic waste is a product of trade—countries export and import plastic across the world. So, where does the plastic we consume actually end up? For wealthier nations, plastic waste often goes to Southeast Asia. Our work reflects this imbalance of more material entering than existing Southeast Asia, with significant mismanaged global plastic waste ending up in our oceans.

Initially, we planned a flow map to trace plastic waste exports to Southeast Asia and other regional redistribution, but inconsistent data forced us to pivot. Disparate sources reported conflicting trade volumes, making accurate flow mapping impossible. Instead, we created a choropleth map showing 2024 per-capita plastic waste by country, a compromise that sacrificed trade insights for clearer standardized data. The challenge lay in reconciling mismatched datasets with varying methodologies. This taught us to prioritize data quality over ideal design: while choropleths lack flow maps' narrative punch, they offer immediate comparability. Key lessons included verifying data uniformity early, embracing flexibility, and, when primary visuals fail, simpler representations can still convey core truths. We also learned to annotate limitations transparently (e.g., "trade data estimated") to maintain credibility. Ultimately, constraints sharpened our focus on Southeast Asia's disproportionate burden, even without granular import/export flows.

Our map exposes Southeast Asia's plastic crisis, where rapid urbanization and weak waste systems collide with global exploitation. Though regional per-capita plastic use is moderate, the area drowns in waste due to inadequate infrastructure and Western exports (8.3% of total waste). With 220M tons of global plastic waste in 2024, 30% leaks into ecosystems, and Southeast Asia bears the brunt: 46.47% of its waste is mismanaged (7.15M tons). ***SDG 12 demands urgent action: wealthy nations must halt waste colonialism, corporations must reduce single-use plastics, and local governments must invest in recycling. The takeaway? This is not just Southeast Asia's problem, it is a global equity crisis.*** Without systemic change, pollution will keep flowing downstream and circulating globally to the world's most vulnerable.

SDG 12 - Responsible Consumption & Production

A closer look at plastic waste in South East Asia

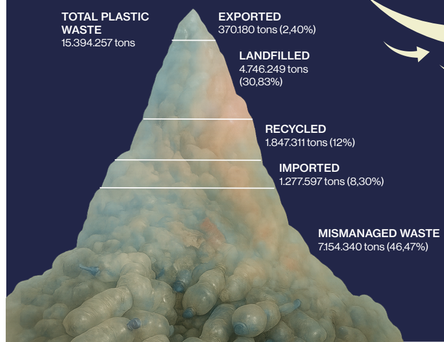
Since the invention of plastic in 1907, its production has exponentially increased globally. At the current rate, yearly global production will exceed 1.3 billion tons by 2060.

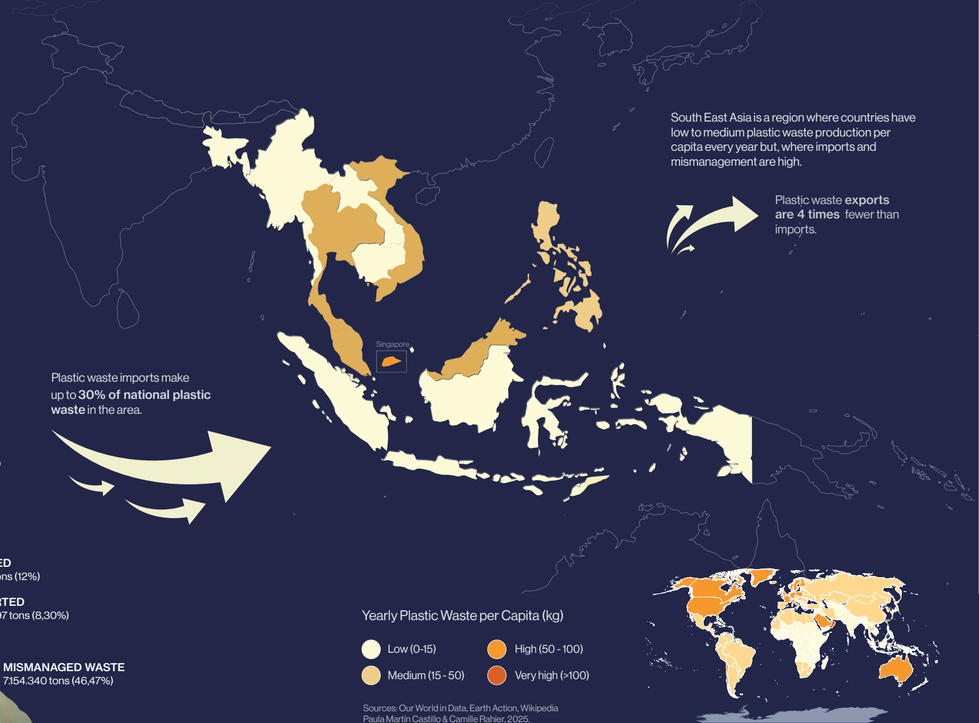
With plastic production, comes plastic waste. In 2024, it is expected that 220 million tons of plastic waste will be generated worldwide.

Plastic waste mainly consists of packaging and construction materials. Everywhere, there is an imbalance between the amount of plastic produced and the ability to manage it. 70% of global plastic waste is managed by landfill (40%), incineration(10%), and recycling (10%).

The remaining 30% is referred as mismanaged waste which ends up in our oceans. This is from the lack of capacity and infrastructure to deal with the waste.

Plastic waste is an item of trade globally, where countries export as much as 60% of their national production. This becomes a problem when it ends up in countries with low waste management capacities.





Sources: Our World in Data, Earth Action, Wikipedia
Paula Martín Castillo & Camille Rahier, 2025.

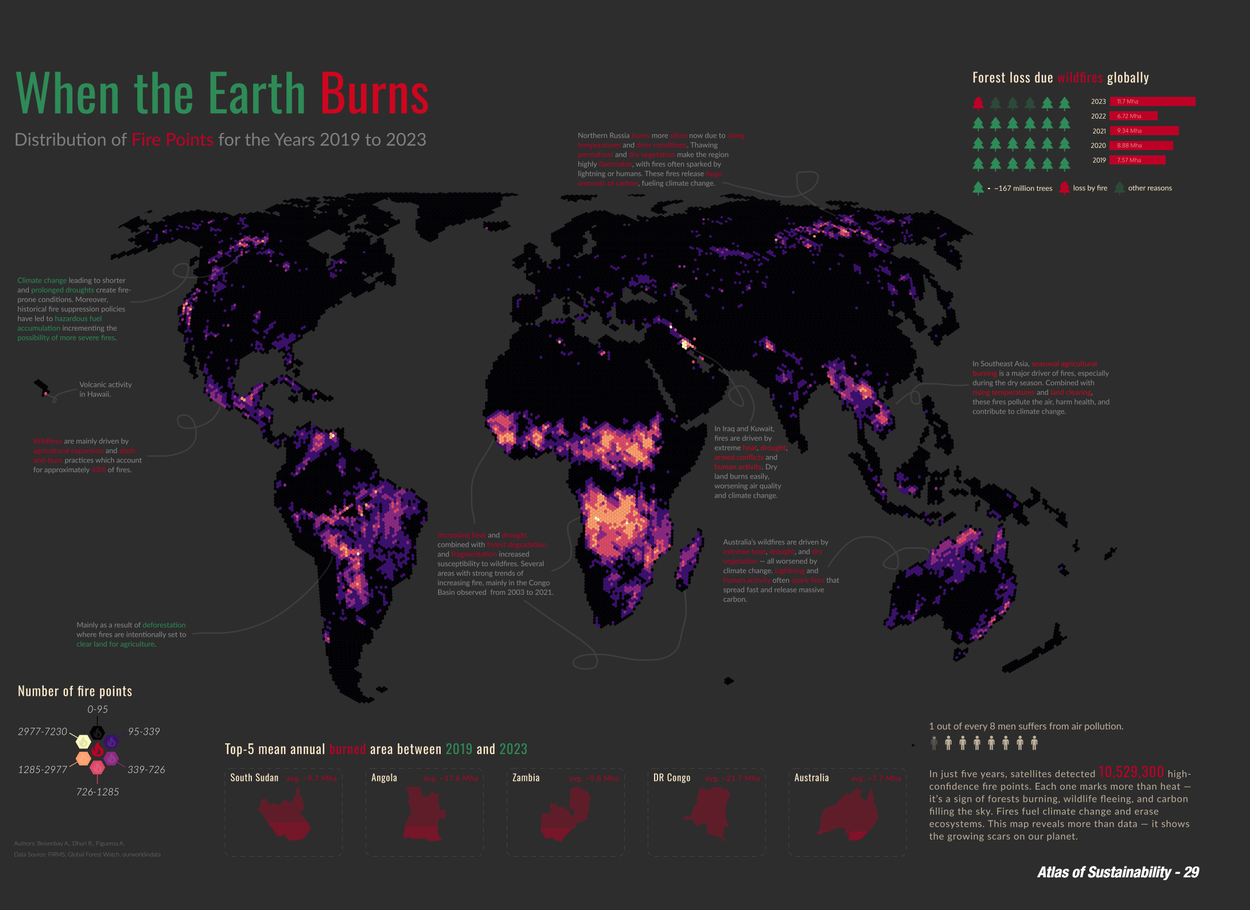
When the Earth Burns

Alikhan Beisenbayev, Radhika Mahesh Dhuri, & Amilcar Figueroa

Sustainable Development Goal 13 on Climate Action is one of the most challenging and pressing SDGs . Our team selected SDG 13 as a top choice as we believe that there is a critical need to actively address the increasing impacts of climate change. The increase in frequency of climate related disasters including natural disasters, rising sea levels, and global warming are all threats to vulnerable populations around the world. We recognize the magnitude of the challenges that are present in this context but also recognize that we can contribute—even if it is it is a small part—our skillsets as cartographers to help address these issues. Our team wanted to approach SDG 13 through a practical perspective by focusing on one specific disaster: fires. By producing a map that shows the occurrence of fire activity around the world, we aimed to communicate which areas were most vulnerable to fires.

For the design process, we drafted a sketch to have an idea of the most important elements where we should focus more. These included the maps we wanted to include and also other graphics like charts or infographics. Eventually, as the process progressed, we got to see which of the elements would have been better suited to the final layout and which elements were dispensable. Working with the actual data and its processing provided us with the clear limitations of what could be displayed as a map and what would be best displayed in other graphics. Moreover, this was a very important case whereby we learned that even our best layout design on the initial stages would have to significantly change as we progressed. As such, our experience highlights the importance of balancing efforts in the process and to no place all efforts of having a clear design layout from the initial stages. In addition, it also shows that designing a map is closely related to a very clear understanding of a dataset, including its limitations and its strengths.

Climate action is the first step towards securing a sustainable future. The map we produced shows the areas where fires have a very dense occurrence in recent years. I***t provides a very important message that tells us that not all areas or countries are equally affected by the effects of climate change. However, this does not mean that only affected countries or regions should act but rather highlights the importance of international collaboration in climate action.*** Moreover, the critical message is that action is always better than words. If no actions are taken, areas affected will keep increasing, the effects will keep growing and the threat to life will become more imminent.



Protecting the Blue: Then and Now

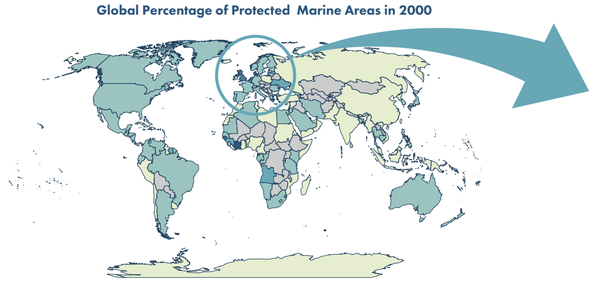
Jennyfer Samaniego Cruz & Devita Remala Sari

Our team selected ***Sustainable Development Goal 14 on Life Below Water*** because we believe ocean and marine conservation is essential for global ecological balance. This SDG focuses on conserving and sustainably using oceans, seas, and marine resources. It includes ten targets to be achieved by 2030, each measured by specific indicators. From the beginning of the Design Challenge, we were particularly drawn to Target 14.5, which aims to conserve at least 10% of coastal and marine areas. We found this target compelling because it is both measurable and can be visualized spatially. This target includes the creation of Marine Protected Areas (MPAs)—zones like marine parks and no-fishing areas where human activity is limited to protect marine ecosystems and biodiversity. Our approach began with exploring how MPAs have expanded over time and how different countries have contributed to this progress. We wanted to use maps and visual storytelling to highlight achievements, particularly in Europe, by comparing the extent of marine protected areas in 2000 and 2023.

To begin our mapping process, we decided to focus on Sustainable Development Goal Target 14.5: Conserve Coastal and Marine Areas. This target was a good fit because the data for its primary indicator was available and suitable for mapping. We started by preparing and generalizing the data related to Marine Protected Areas (MPAs). After uploading the dataset into ArcGIS Pro, we explored patterns and outliers in the spatial distribution and discussed possible design directions. As a first step, we chose to create a world map highlighting the countries with the highest percentage of protected marine areas in 2023. From there, we decided to focus specifically on Europe, as the continent has shown the most significant progress in marine protection over the last 23 years. Between 2000 and 2023, many European countries have notably increased the proportion of their marine areas under protection, making it a strong regional case for illustrating change over time.

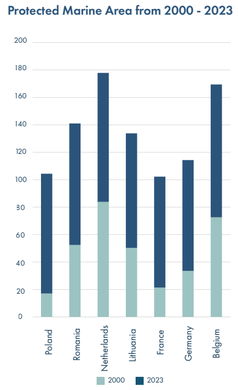
Our final design tells the story of how marine protection has improved over time, with a focus on Europe. Starting with Sustainable Development Goal 14.5, we wanted to highlight where progress has been made towards conserving at least 10% of marine and coastal areas. We chose to compare the years 2000 and 2023 because this time span clearly shows how much progress has been made in expanding protection. Our map does not show the exact protected zones, but instead highlights the proportion of marine protection at the country level, helping us compare which countries have done more. Europe stood out because many of its countries more than doubled their coverage, showing regional leadership in marine conservation. ***Our main message is that while progress is happening, especially in certain regions, more effort is needed worldwide to meet these goals and protect our ocean ecosystems.***

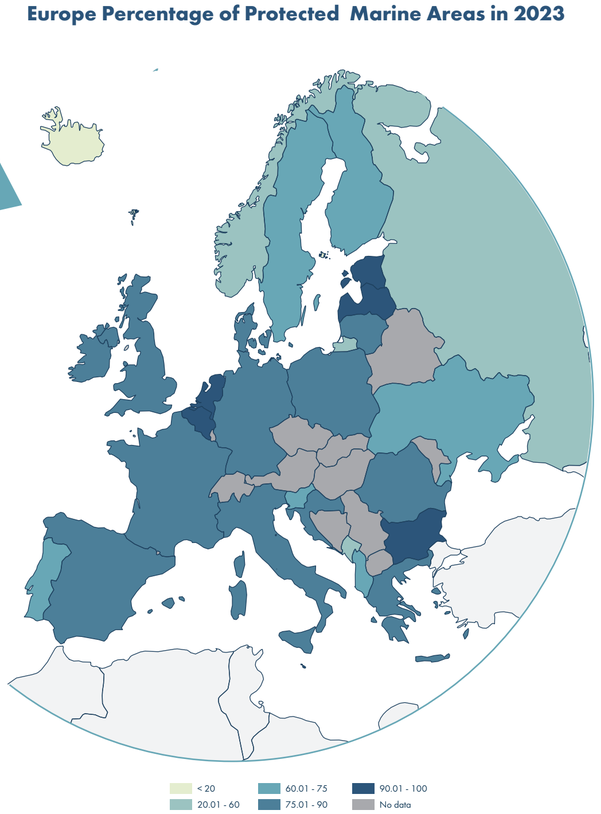
Protecting the Blue:
Then & Now



Between 2000 and 2023, marine and coastal protected areas in Europe have grown significantly, reflecting major efforts by EU Member States to safeguard their marine environments. Countries such as Germany, Belgium, and France have each designated more than 30% of their marine waters as Marine Protected Areas (MPAs), while others, including the Netherlands, Lithuania, Poland, and Romania, have expanded their MPA networks to cover over 20%.

This map illustrates the progress made across the continent by showing the proportion of marine areas protected within each country, rather than the exact locations of the protected areas themselves. It highlights how various countries have expanded the coverage of their marine protection efforts over time.





15 LIFE ON LAND

Species on the Edge: Red List Endangered Animals

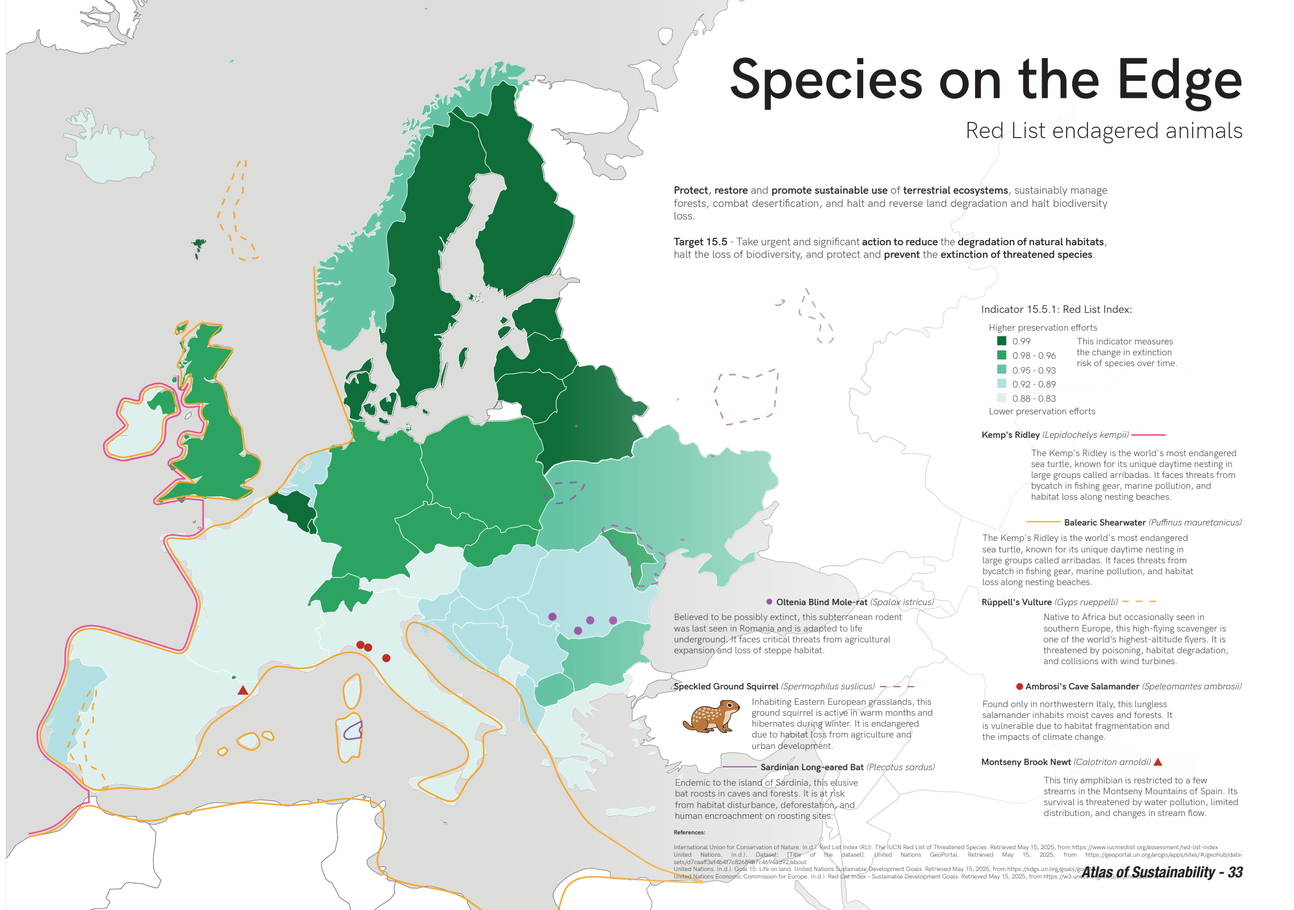
Pedro Bonilla Artigas, Yi-Ru Liang, & Tim Pauly

Our team focuses on ***Sustainable Development Goal 15 about Life on Land***, which aims to protect, restore, and promote the sustainable use of terrestrial ecosystems. We were particularly drawn to Target 15.5, which calls for urgent action to halt the loss of biodiversity and protect threatened species from extinction. We selected this goal because we all care about biodiversity and wildlife conservation. At the beginning of the Design Challenge, we wanted to highlight the urgency of biodiversity loss in a way that was both informative and emotionally impactful. To approach this, we decided to create a basemap of Europe that combines data from the Red List Index, showing general biodiversity health, with habitat data for eight endangered species. This way, we could visualize both the big-picture in biodiversity and tell individual stories of species at risk.

During the event, our design process began with exploring indicator datasets inside the SDG global database. Initially, we tried to review as much data as possible to get inspiration, but this approach was overwhelming and did not lead to a clear direction. The turning point came when we discovered the Red List species database on the IUCN website. This helped us narrow our focus to endangered species and connect our project more directly to SDG Target 15.5. From there, we spent some time thinking about how to visualize species habitats.

We discussed various options, and we decided to use a combination of points and lines to represent where each of the eight selected species live.

As we worked on this project, we realized that the situation with biodiversity in Europe is not entirely negative. While some species are critically endangered, there are also countries making real efforts to protect ecosystems. Our map tells a story of how biodiversity health varies across Europe, and how some places are doing better than others when it comes to conservation. At first, we were just interested in animals and thought we should explore general data, but it did not feel very inspired until we discovered the IUCN Red List and started connecting it with actual species. That helped us shift our focus to a more visual comparison—using the Red List Index to show overall biodiversity health by country, and overlaying it with the habitats of eight endangered species. We noticed that many of these habitats fall in regions with lower Red List indexes, which made us think about the relationship between national conservation efforts and species survival. ***Our takeaway is that species loss is not just happening faraway; it is happening right here in Europe and to animals most of us rarely think about.*** We hope our map lets people see where wildlife conservation is working and where more attention is needed.



Strategic Partners or New Masters? Russian and Chinese Influence in Africa

Anna Shelegina & Anthony (Ching Yeung) Wong

Our team chose ***Sustainable Development Goal 16 on Peace, Justice, and Strong Institutions*** because it is often overlooked among the SDGs due to its abstract nature and sensitivity around topics like governance, conflict, and geopolitical influence. Unlike more tangible goals such as clean water or quality education, SDG 16 requires engaging with power structures and asking difficult questions about accountability and justice. In the 2024 Global Progress Report, many of SDG 16's indicators were shown to have stagnated or declined, but there was little exploration into why this was happening. While the war in Ukraine has drawn global attention and is cited as a major driver of instability, we wanted to shift the lens to Africa, where conflict and institutional weakness are often tied to deeper economic and geopolitical interests. We asked: what role do foreign powers play in shaping these outcomes? With rising Chinese infrastructure projects and growing Russian military involvement, we saw a pattern of partnerships that may not always benefit local populations. Our project began with a desire to explore whether these forms of "collaboration" reflect genuine development—or if they mask a modern form of neo-colonialism that ultimately undermines the principles of peace, justice, and institutional strength that SDG 16 aims to promote.

The hardest part was making an unusual binomial map. We spent a long time choosing the color palette, but ultimately did not find an optimal scheme. In the end, we spend a couple of hours at night to finally come to the relief overlay that makes the map more alive. We also wanted to avoid colors that are associated with China, Russia, war, etc. Throughout the process, we learned to work quickly and not get stuck on one idea.

Our final design visualizes how foreign influence by China and Russia in Africa threatens the achievement of SDG 16 on Peace, Justice, and Strong Institutions. Using a bivariate map, we reveal patterns of strategic agreements, port control, and military presence—particularly where China has invested heavily in infrastructure and Russia has embedded private military contractors like the Wagner Group. While these partnerships are often framed as developmental or stabilizing, our data shows that they frequently serve the strategic interests of the foreign powers, not the local populations. For example, Chinese contracts often import their own labour (up to 70%) and operate under opaque terms, limiting economic justice and transparency. Similarly, Russian military support tends to reinforce authoritarian regimes, exacerbating conflict and undermining the rule of law. The key challenge is that these foreign powers are shaping African governance structures from the outside, weakening public accountability, inflaming corruption, and sidelining inclusive decision-making. ***Our conclusion is clear: SDG 16 cannot be realized in regions where sovereignty is compromised and institutions are captured by external agendas.*** The map acts as both a diagnostic tool and a warning—highlighting where urgent attention is needed to preserve justice, peace, and truly strong institutions.

STRATEGIC PARTNERS or NEW MASTERS?

Unpacking Russian and Chinese Influence in Africa

Russia and China have **expanded** their presence in Africa through **military contracts** and **port investments.**While framed as development or security cooperation, these relationships often **serve external strategic or economic interests** —at the expense of African sovereignty, justice systems, and accountable governance.

2025 — **Russia signs new Military Agreement with Burkina Faso**

2023 — **Military coup in Niger**

2022 — **China and Russia forge "no limits" strategic pact**

2021 — **China establishes presence in Djibouti, Ethiopia, Mali**
Military coup in Mali

2019 — **Coup in Sudan**

2014 — **Post-Crimea: Russia shifts focus to Africa**
Russian presence begins in Central African Republic

2011 — **China launches Belt and Road Initiative**

2006 — **China announces $8B in African development loans**

2000 — **China-Africa Cooperation Forum (FOCAC) launched**

1993 — **China-Africa trade agreement**

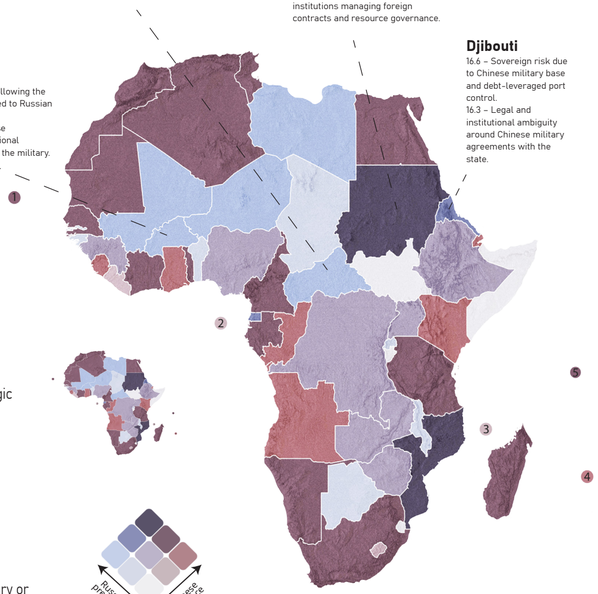
1990 — **Wagner Group established**

Central African Republic
16.1 – Escalation of violence and conflict (Wagner Group presence).
16.3 – Undermining the rule of law by empowering non-transparent, foreign-controlled security operations.

Sudan
16.5 – Corruption linked to secretive gold extraction deals (Russia) and port projects (China).
16.6 – Lack of accountable institutions managing foreign contracts and resource governance.

Djibouti
16.6 – Sovereign risk due to Chinese military base and debt-leveraged port control.
16.3 – Legal and institutional ambiguity around Chinese military agreements with the state.

Burkina Faso
16.1 – Increased military cooperation following the 2022 coup, including deployments linked to Russian private military groups.
16.6 – Weak oversight of foreign defense agreements risks undermining institutional accountability and civilian control over the military.



Color reflects
Chinese presence: ports, infrastructure, strategic deals
Russian presence: military agreements, troop deployment, milirary bases

Scoring system:
0-No involvment
1-Strategic agreement (infrustructure or security)
3-Strategic and active military or economic control

17 PARTNERSHIPS FOR THE GOALS

Data Delivered? Reporting SDG Tier 1 Indicators in 2020

Anna Auermüllerová, Gabriela Godišková, & Dajana Snopková

For our Design Challenge project, our team chose to focus on ***Sustainable Development Goal 17 on Partnerships for the Goals***. We selected this SDG because we believe that effective global cooperation and transparent reporting are fundamental to achieving all other development goals. At the start of the project, we wanted to explore how actively different countries engage with the SDGs through official reporting. Our idea was to create a global map that visually represents the extent to which countries are fulfilling their reporting duties. To do this, we gathered data showing which countries submitted voluntary national reviews (VNRs) and how many of the SDGs sub-targets they addressed. We then calculated the percentage of sub-targets reflected in each country's report. This allowed us to build a dataset showing not just whether a country participated, but how comprehensively it responded. Our plan was to visualize this percentage using a color gradient on a world map, where each country's color indicates the proportion of SDGs targets it addressed in its report. We hoped this approach would clearly communicate differences in engagement and provide insight into global cooperation efforts in sustainable development.

After we selected our SDG goal, we started researching the topic. The goal itself is really broad, so we were trying to narrow our focus and tried to look for one specific topic. We saw our goal as a parent to the other 16 goals, so after careful consideration, we decided to have a look at how members are providing data about their other goals' fulfilment. Data were available through the API on the United Nations Statistics website. Using Postman, we conducted a dataset of 98 Tier 1 indicators. While the data were being retrieved, we also were sketching our first design of the layout. As we moved from GIS to Adobe Illustrator, we realized that our data seemed biased by alphabetical order. Eventually, we figured that our API call was only collecting the first "part" for each indicator. We edited the call and retrieved our data once again, this time capturing the entire database. Using the complete dataset, we created a map visualization showing the percentage of reported T1 indicators per member country. Our experience is a useful reminder the maps can help identify errors in data collection in processing as well as tell compelling visual stories!

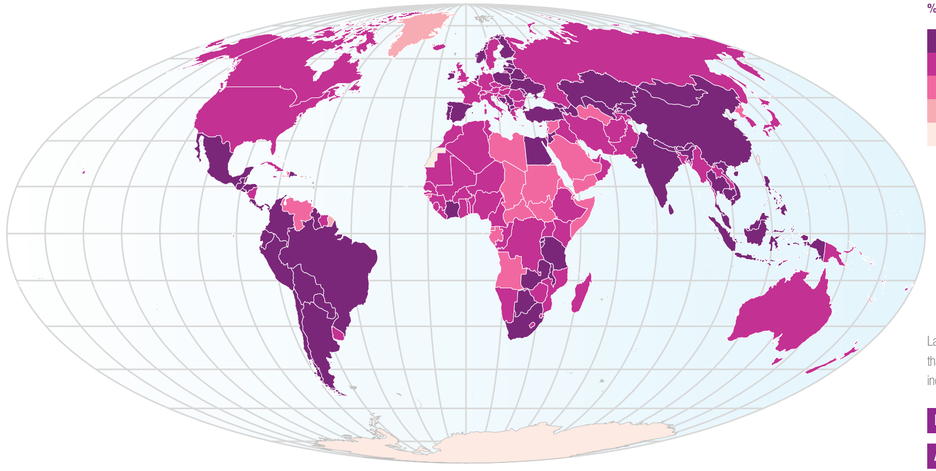
SDG 17, Partnerships for the Goals, combines cooperation and implementation of the remaining 16 goals. The level of success in reaching a specific goal is measurable through indicators—each goal has its own set of measures. Indicators are divided into classified into tiers that specify the comprehensiveness in indicator reporting, with Tier 1 (T1) indicatorsreported by a majority of United Nations member states on a regular basis. When providing information about the state of any SDG achievement, transparency is the key element. The rate of indicator publication demonstrates such transparency in communicating a country's SDG achievements. This map shows the percentage of reported T1 indicators per country in the year of 2020 - the higher the percentage, the more indicators were reported among all 98 T1 indicators.

It is visible that there are many countries reporting around three-quarters of all T1 indicators. The highest rates are reached within Latin America (countries providing more than 80% of the indicators). Further, almost all member countries are providing information about more than 40% of the indicators, reaching nearly half of the total number of T1 indicators. Disclaimer: When interpreting such data, one must be aware that besides transparency and regular indicator reporting, there must be reliability of the numbers reported. It is therefore recommended to explore additional sources.

Data Delivered?

How did the UN reporters deliver all of their SDG Tier 1 (T1) indicators in 2020?

The map below shows the percentage of indicators reported among all 98 T1 indicators for each country.



Latin America as a Leader

Latin America is a home to the countries that delivered data about more than 80% T1 indicators.

Mexico	84%
Argentina	83%
Peru	81%

Data were obtained by using API call https://unstats.un.org/SDGAPI/v1/sdg/Indicator/Data?indicator=[indicator_code]&timePeriod=2020

Afterword: *TITLE*

Georg Gartner, President, International Cartographic Association

The ***University of Wisconsin Cartography Lab (Cart Lab) Design Challenge*** is a day-long mapping workshop that brings together Cartography and Geography students, often in collaboration with campus or community partners, around a curated mapping theme. The Design Challenge is now an institution in the University of Wisconsin–Madison Geography Department, first running in 2015 and wrapping its 10th annual event this past February 2024. This year (officially #DC11) we brought the Design Challenge to an international stage, combining efforts with the ***Technical University of Wien Cartography Research Unit***, the ***Erasmus Mundus MSc Cartography Programme***, and the ***International Cartographic Association Commission on the User Experience (UX)***.

Specifically, our respective groups came together to make a collective atlas that tells regional and local stories about the United Nations Sustainable Development Goals (SDGs) ***... in just one week!*** For background, the United Nations adopted the 2030 Agenda for Sustainable Development in 2015, an expansion from the prior 2000–2015 Millennium Development Goals, and defined a global indicator framework for collective transformation and action toward a more equitable and just world. At the time of this mapping workshop (May 2025), the SDG global indicator framework comprised 17 goals to address the most pressing problems facing our worlds, 169 targets that prescribe real and actionable outcomes for sustainability, and 231 unique indicators used to measure and monitor progress towards these targets, inform policy at global and local levels, and promote accountability across all stakeholders (https://unstats.un.org/sdgs/).

The 2025 event, held in Wien May 12-16, built on prior collaborations among the International Cartographic Association, the United Nations, and the UW Cart Lab to develop the ***Mapping for a Sustainable World*** (Kraak et al. 2020) open source Cartography textbook and subsequent spin-off activities to create the ***QGIS Technical Supplement*** (Houtman et al. 2021) supporting mapping of the SDG global indicator framework. The 2025 event was an overwhemling success, demonstrating the promise and possibilities of international collaboration. The resulting ***Atlas of Sustainability*** follows the ***Guerilla Cartography*** "Atlas" model (Cowart et al. 2024) and includes 17 two-page spreads, each covering an SDG, plus some front and back matter. Each spread includes one page for maps, charts, and clarifying annotation about the selected SDG and a second page providing a title and statement about what the maps are saying and what this means for our planet's future and our individual actions.

MSc Cartography students along with visitors from Masaryk University (Brno) worked in small teams to develop one of the atlas spreads. As part of mapping SDG indicator data, we also asked the international students to bring their own personal experiences and home locations into the map, resulting in 17 compelling stories grounded from situated perspectives to explore the alternatives that are possible from their different, worldwide perspectives. The #DC11 schedule was broken into six 90-120 minute units spanning three days, with most scheduled activities on Monday, May 12th, and drafts submitted 24 hours after the final unit on Thursday, May 15th. We aseembled in its presented form on Friday, May 16th. As an ongoing project, we invite you all to join us in mapping for a sustainable world...the ***Atlas***.